From credit-risk model to governed AI system

An end-to-end demonstrator covering development, validation, explainability, fairness, monitoring, and governance.

Data Governance | Baseline | Challengers | Explainability | Model Risk | Monitoring + Governance

Model selection and test performance

PR-AUC
0.5669
Default ranking

BRIER SCORE
0.1332
Probability accuracy

Why the XGBoost result is credible

- ROC-AUC improvement vs Logistic Regression: +0.0263.
- Paired-bootstrap 95% CI for AUC gain: +0.0168 to +0.0369.
- Calibration slope: 0.9460. Champion selected before final test review.
- Test set remained untouched until model family, calibration, and threshold were frozen.

Explainability and Responsible AI

EXPLAINABILITY

0.9972

Median SHAP rank Spearman

FAIRNESS

5 audits

SEX, AGE, EDUCATION, MARRIAGE, intersection

HUMAN OVERSIGHT

8.09%

Review-band volume

- TreeSHAP global and local explanations with reason-code mapping.
- Demographic attributes excluded from training and retained for audit only.
- SEX alone did not trigger the internal screening threshold.
- AGE, EDUCATION, MARRIAGE, and SEX x AGE remain monitoring questions.

Model-risk validation

Challenge	Finding	Governance interpretation
Reproducibility	Max probability diff 2.98e-08	**PASS**
Deduplicated training	AUC change -0.0013	**Stable**
Repayment-code recoding	AUC change -0.0009	**Stable**
Winsorization	AUC change +0.0004	**Stable**
Remove repayment history	AUC change -0.0510	**Critical dependency**
External validity	Historical Taiwan data	**High residual risk**

Validation conclusion

- The champion is robust to several data-treatment choices, but repayment history is a material dependency.
- External validity remains open. Contemporary institution-specific data is required before production use.

Production-style monitoring snapshot

GREEN KRIS	AMBER KRIS	RED KRIS	NOT AVAILABLE
14	4	0	2
Current backtest status	Current backtest status	Current backtest status	Current backtest status

Monitoring controls

- Max critical-feature PSI: 0.0043 | Score-distribution PSI: 0.0055
- Review-band volume: 8.09% | Fairness AMBERs remain recurring KRIs
- Override-rate KRIs stay NOT_AVAILABLE because no reviewer logs exist
- Pseudo-period alerting demonstrates how weaker batches trigger investigation

Governance pack

AI risks	12
Governance controls	17
RACI activities	17
Regulatory mappings	15
Governance visuals	6
Integrity validation	PASS

Enterprise governance architecture

1st Line

Business + Credit Risk

Model Owner + Operations

2nd Line

Model Risk + AI Governance

Compliance + Data Governance

3rd Line

Internal Audit

Independent Assurance

Portfolio value and scope

- Connects credit-risk modelling with AI governance and model-risk management.
- Maps evidence to selected EU AI Act, GDPR, EBA, and ECB governance themes.
- Demonstrates a lifecycle from data governance to monitoring and escalation.

Important limitation

Not production ready. No IRB, Basel, or compliance-certification claim. Contemporary institution-specific validation is required.

github.com/naveeddatar/governed-credit-ai-governance